

Oblique water entry of a three dimensional body

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ABSTRACT: *The problem of the oblique water entry of a three dimensional body is considered. Wagner theory is the theoretical framework. Applications are discussed for an elliptic paraboloid entering an initially flat free surface. A dedicated experimental campaign yields a data base for comparisons. In the present analysis, pressure, force and dynamics of the wetted surface expansion are assessed.*

KEY WORDS: Wagner problem; Oblique entry; Elliptic paraboloid.

INTRODUCTION

This study aims at improving the numerical Tools for dimensioning the Ultra Large Container Ships (TULCS). This is a European project of the Seventh Framework Program with grant agreement number FP7-234146. Slamming loads are parts of the hydrodynamic loads acting on the hull of container ships; mainly on their bow part. Actual configurations are difficult to handle. However we are here concerned with the early stages of penetration of such bodies in water. Therefore simplifications are possible. One of them is to reduce the geometrical definition of the body shape by using its two radii of curvature at the initial contact point.

The oblique water entry of an elliptic paraboloid is hence considered in the frame of the linearized Wagner problem. Basically the boundary value problem is posed in Potential Theory for the displacement potential which is the time integral of the velocity potential. As long as the wetted surface is elliptic, the Galin's theorem (see Galin, 1953) provides that displacement potential. We focus on the oblique entry and it is shown that this configuration slightly differs from the pure vertical entry. As a consequence the leading order vertical force remains unchanged whatever the horizontal kinematics. However the local loads are substantially modified. In particular, the appearance of negative pressure is increased as the horizontal velocity increases. The pressure formulation is improved by using the Modified Logvinovich Model (MLM). Comparisons are made with experimental data. The experimental program is performed in BGO First (La Seyne/mer, France). Pressures, forces and time variations of the expanding wetted surface are analyzed and compared to the theoretical results. The agreement shows that Wagner model is quite satisfactory.

The arrangement of the paper is as follows. The first section describes the theoretical developments. The solution for penetration with two Degrees of Freedom (DoF) is detailed. The calculations of global and local loads are described in a second section. The comparisons of theoretical and experimental data are discussed in the third section. In appendix, the Galin's theorem and its application for solving the present problem are detailed.

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THE LINEARIZED WAGNER MODEL

The linearized Wagner problem is formulated in Potential Theory in the absence of gravity and surface tension and under the assumptions of the so-called flat-disk approximation (see Korobkin and Pukhnachov, 1988). That means that the body is flat enough and this flatness is measured by the angle between the body and the fluid surface. We study here an elliptic paraboloid. The initial dead rise angle is thus zero when the body hits the liquid flat free surface. Air cushion effects may hence occur depending of the importance of the curvature radii. Those effects are not accounted for in the present approach.

Method of solution

It is well known that the three dimensional Wagner problem applied to arbitrary shapes is still open (see Scolan and Korobkin 2008). However exact solutions exist when the body shape is simple. In Scolan and Korobkin (2001) a survey collects the different methods of solution applied to three dimensional bodies impacting a flat free surface. In particular the inverse problem yields exact solutions under the assumption that the flat disk (the linearized wetted surface) is elliptic whatever the slope at the initial contact point: discontinuous for an elliptic cone or continuous for an elliptic paraboloid. For the latter shape the inverse problem can be revisited by using the Galin's theorem which appears to be a powerful shortcut. To this end, the Boundary Value Problem (BVP) is formulated with the displacement potential. This potential is the time integral of the classical velocity potential. The corresponding Boundary Value Problem is formulated in Korobkin and Scolan (2006) and it reads

$$\begin{cases} \phi_{,xx} + \phi_{,yy} + \phi_{,zz} = 0 & z < 0 \\ \phi = 0 & z = 0, (x, y) \in FS(t) \\ \phi_{,z} = -h(t) + f(x, y) & z = 0, (x, y) \in D(t) \\ \phi \rightarrow 0 & (x^2 + y^2 + z^2) \rightarrow \infty \end{cases} \quad (1)$$

where the regions $FS(t)$ and $D(t)$ are disconnected parts of the plane $z = 0$ and correspond to the free surface and the wetted area of the body, respectively. The closed curve, which separates the regions $FS(t)$ and $D(t)$, is denoted $\Gamma(t)$ and it is referred to as the contact line. The Neumann condition on $D(t)$ depends on the penetration depth h and on the shape function $z = f(x, y)$ of the penetrating body. The origin of the coordinate system in which the shape is defined is centered at the initial contact point. From the BVP (1) we derive an integral equation

$$\frac{1}{2\pi} \iint_{D(t)} \frac{S(x_0, y_0, t) dx_0 dy_0}{\sqrt{(x-x_0)^2 + (y-y_0)^2}} = h(t) - f(x, y) \quad (x, y) \in D(t) \quad (2)$$

where S denotes the planar Laplacian of ϕ hence $S(x, y) = \phi_{,xx} + \phi_{,yy}$. It is worth mentioning that according to Stephan (1987) the planar Laplacian S has a square root singularity at the contact line. By solving the integral Eq. (2), we obtain a Poisson equation for ϕ which satisfies the following boundary condition

$$\phi = 0 \quad \text{on } \Gamma(t) \quad (3)$$

An additional condition is prescribed along the contact line. This condition implies that not only the displacement potential $\phi(x, y, 0, t)$ and the vertical displacement $\phi_{,z}(x, y, 0, t)$ but also the horizontal displacements $\phi_{,x}(x, y, 0, t)$ and $\phi_{,y}(x, y, 0, t)$ are continuous through $\Gamma(t)$. With account for (3) the latter condition can be presented as

$$\phi_n = 0 \quad \text{on } \Gamma(t) \quad (4)$$

where $\phi_n = \phi_{,x}(x, y, 0, t)n_x + \phi_{,y}(x, y, 0, t)n_y$ and $n = (n_x, n_y)$ is the unit normal vector along the contact line $\Gamma(t)$.

The solution of the integral Eq. (2) is given by Galin (1953) provided that the support of integration $D(t)$ is elliptic and the shape function f is a polynomial. Details about Galin's theorem and its application are given in appendix. As a consequence it is shown that an elliptic paraboloid entering a flat free surface with an arbitrary vertical kinematics has an elliptic wetted surface at any time. Another proof of the latter statement is given by Korobkin (2002). We denote (a, b) the semi axes of the elliptic wetted surface along the x and y directions respectively. Without any angular motions of the entering body, it is also shown that neither the aspect ratio $k = a/b$ nor the eccentricity $e^2 = 1 - k^2$ depend on time. As shown in Eq. (29) of the appendix, both a and b vary as the square root of the penetration depth h . That yields the following identities

$$2h\dot{a} = h\dot{a}, \quad \frac{\dot{a}}{a} = \frac{\dot{b}}{b} \quad (5)$$

and the displacement potential reads

$$\phi(x, y, t) = -\frac{2ha}{3E} \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} \right)^{\frac{2}{3}} \quad (6)$$

where E is a standard Elliptic Integral (see appendix).

If the coordinate system (x, y) is non inertial, -in other words the coordinate system (x, y) is attached to earth- then, by using the identities (5) and denoting the vertical velocity $W = \dot{h}$, the expression of the velocity potential (noted φ) is

$$\varphi = \frac{d\phi}{dt} = -\frac{aW}{E} \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} \quad (7)$$

Entry with more than one DoF

The case of the elliptic paraboloid which enters an initially flat free surface with more than one DoF is fully described in Scolan and Korobkin (2014). Fig. 1 shows the configuration.

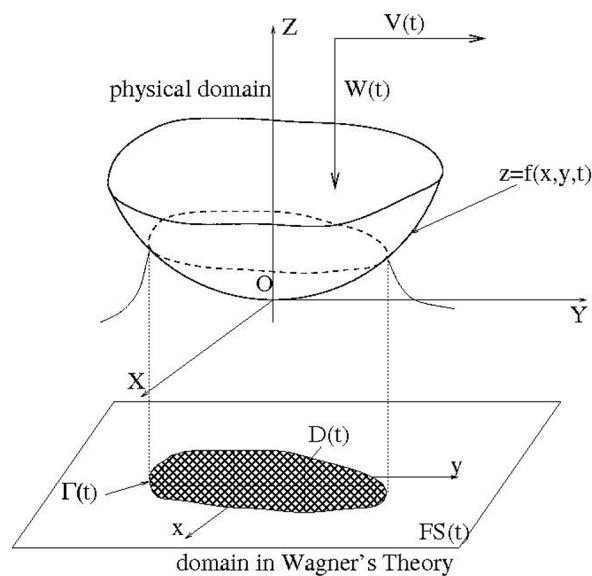


Fig. 1 Physical configuration of a three dimensional body entering an initially flat free surface with vertical and horizontal velocities. The lower sketch illustrates the linearization made in the Wagner's model.

The restriction to the combination of vertical and horizontal translational motions is treated in the next developments. The horizontal motion occurs in the y direction. We consider that the elliptic paraboloid is described in a local coordinate system (X, Y, Z) by the equation $Z = f(X, Y)$ where the function f is given by (23). We denote $(\delta y, -h)$ the instantaneous position of the lowest point $(X, Y) = (0, 0)$ in a coordinate system attached to earth. The displacement of any point on the surface of elliptic paraboloid then reads

$$x = X, \quad y = \delta y + Y, \quad z = -h + Z \quad (8)$$

We use the change of variables, $\tilde{y} = y - \delta y$ so that the vertical displacement which appears in the Neumann condition on the wetted surface in BVP (1) becomes

$$\phi_{,z} = -h(t) + f(x, \tilde{y}) \quad (9)$$

while the other equations remain unchanged but now in the coordinates system (x, y, z) . The method of solution yields the same solution and the displacement potential reads

$$\phi(x, \tilde{y}, t) = -\frac{2ha}{3E} \left(1 - \frac{x^2}{a^2} - \frac{\tilde{y}^2}{b^2} \right)^{\frac{2}{3}} \quad (10)$$

We conclude that the size of the wetted surface in the coordinates $(x, \tilde{y})$ is not affected by the horizontal velocity of the penetrating body and the formulae (29) still provide the two semi-axes (a, b) . From the displacement potential (10) we derive the velocity potential φ

$$\varphi(x, \tilde{y}, t) = -\frac{a}{E} \left(W + \frac{2hV\tilde{y}}{b^2} \right) \sqrt{1 - \frac{x^2}{a^2} - \frac{\tilde{y}^2}{b^2}} \quad (11)$$

Eq. (11) shows that the velocity potential vanishes at the contact line as the square root of the distance to the contact line. Eq. (11) reduces to Eq. (4.10) of Scolan and Korobkin (2001) if the horizontal velocity is zero. However, in contrast to the pure vertical kinematics, the expression of the velocity potential (11) does not reduce to the product of the square root of the contact line function times a function of time only, as suggested by Leonov (1940). Hence the models based on that assumption fails at solving properly the oblique entry problem (see Tassin et al., 2012).

GLOBAL AND LOCAL LOADS

Pressure

At the leading order, the pressure follows from the linearized Bernoulli equation

$$p = -\rho\varphi_{,t} = -\rho\phi_{,t^2} \quad (12)$$

We consider an elliptic paraboloid which enters the free surface with vertical velocity $W = h_{,t}$ and horizontal velocity $V = \delta y_{,t}$. The spatial and temporal variations of the pressure are analyzed in Scolan and Korobkin (2012a). In particular it is shown when and where the pressure becomes lower than atmospheric pressure on the wetted surface. The so called Modified Logvinovich Model for the pressure (see Korobkin and Malenica, 2005; Korobkin, 2005) is considered as an improvement of

the pressure calculation since we get a uniformly valid distribution up to the contact line where pressure reaches the atmospheric pressure. The equation for the MLM pressure with horizontal velocity reads

$$-\frac{1}{\rho}P(x, y, t) = \varphi_t^{(1)} - \dot{W}(f(x, y - \delta y) - h) + \frac{1}{2}W^2 + \frac{1}{2}\bar{\nabla}^2\varphi^{(1)} - \frac{1}{2D}\left((\bar{\nabla}\varphi^{(1)} - \bar{v}) \cdot \bar{\nabla}f\right)^2, \quad D = 1 + \bar{\nabla}^2f \quad (13)$$

where $\bar{v} = (0, V, 0)$ is the horizontal velocity and $\varphi^{(1)}$ is the linearized velocity potential given by (11).

Force

At the leading order, for small penetration depth, the vertical force $F_z(t) = \vec{F} \cdot \vec{z}$ follows from the integration of the pressure (12) over the contact region $D(t)$. From Newman (1977) we can also formulate the vertical force as

$$F_z(t) = -\rho \frac{d}{dt} \iint_{D(t)} \varphi dx dy \quad (14)$$

where the support of integration is the linearized wetted surface $D(t)$. By using the expression of the velocity potential (11) and the identities (5), the vertical force reads

$$F_z(t) = \frac{4\pi\rho}{15E} k^2 \eta^3 \frac{d^2}{dt^2} [h^{5/2}] \quad (15)$$

where η is given in (29). It is worth noting that, at the leading order, the force depends on the time variation of the penetration depth only since the factor $k^2\eta^3/E$ does not depend on time in the present case. We also note that the force contains two terms: one in phase with the acceleration $h^{3/2}h_{tt}$ and another one varying with the velocity square $h^{1/2}(h_t)^2$. Assuming temporarily that the vertical velocity W is constant, the force reduces to

$$F_z(t) = \frac{1}{E} \pi \rho k^2 \eta^3 W^{5/2} \sqrt{t} \quad (16)$$

At the leading order the horizontal component of the force is zero. However a first order correction can be calculated as follows

$$\vec{F}(t) \cdot \vec{y} = F_y(t) = \iint_{D(t)} \rho \varphi n_y ds \quad (17)$$

where the velocity potential φ is by (11). Knowing that

$$n_y ds = \frac{\tilde{y}}{R_y} dx d\tilde{y}$$

where R_y is the curvature radius along the $\vec{y}$ direction. Then end up with the horizontal force

$$F_y(t) \approx -\frac{4\pi\rho}{15ER_y k^2 \eta^3} \frac{d}{dt} \left(Vh \frac{5}{2} \right) \quad (18)$$

DISCUSSION

In order to validate the theoretical results, we use the experimental data base obtained during the experimental program described in Scolan (2012) and Le Hir (2012). An elliptic paraboloid is set on a forced motion generator above a free surface as illustrated in Fig. 2.

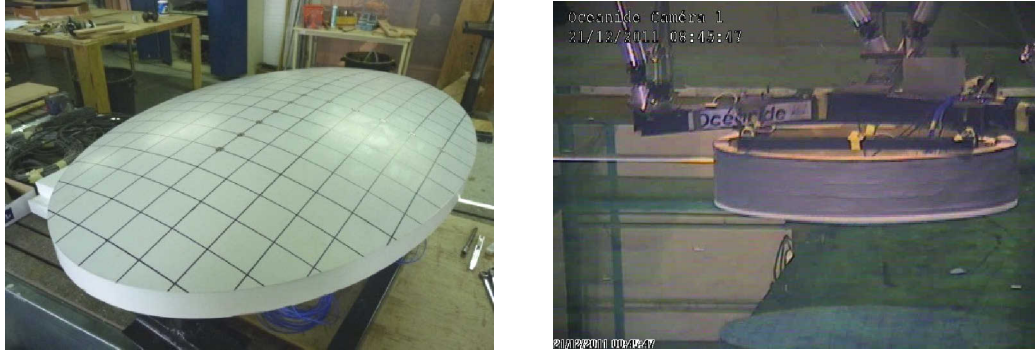


Fig. 2 Experimental set-up in BGO-First (La Seyne/Mer, France), (left): shape of the elliptic paraboloid, (right): shape set on the forced motion generator.

The available motion generator imposes two constraints: 1) a maximum allowed force and 2) a maximum amplitude of velocity (whatever its direction). The amplitude of velocity cannot be greater than 1m/s. From that constraint the largest possible size of the elliptic paraboloid is defined by the two radii of curvature set to $R_x = 0.75 \text{ m}$ and $R_y = 2 \text{ m}$ along the x and y directions. The aspect ratio of the shape is $k_\gamma = 0.61237$ and the aspect ratio of the elliptic contact line is $k = 0.67422$. The instrumentation allows to measure the vertical and horizontal forces. The pressure is also measured at several locations along the symmetry axes of the elliptic paraboloid. Fig. 3 compares the time variation of the vertical force either experimental or theoretical (Eq. (16)) made nondimensioned by $W^{5/2}$.

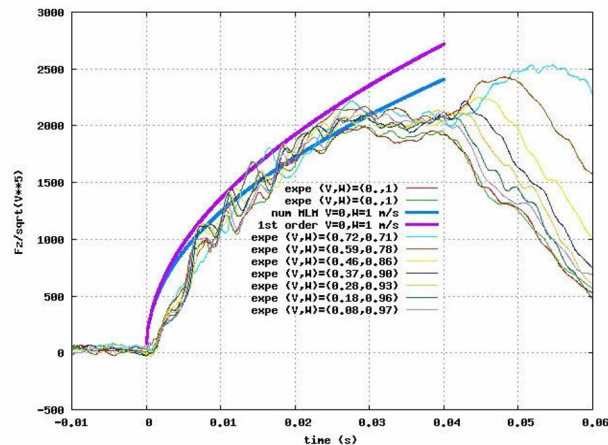


Fig. 3 Time variation of the vertical force. Comparison between theoretical force (Eq. (16)), numerical integration of MLM pressure (Eq. (13)) and experimental data for pure vertical motions and oblique motions.

As usual the first order formulation (16) slightly overestimates the force. As shown in Korobkin (2004) the agreement is improved by numerically integrating the MLM pressure (13) over the wetted surface. That requires to precisely detect the point where the pressure vanishes close to the contact line. Unfortunately the comparison of the measured and computed horizontal force is not possible. The main reason is a lack of stiffness of the setup which introduces perturbations of the same order than the measured force itself.

Regarding the expansion of the wetted surface, we observe the body penetration by using a submerged camera. It records along a vertical axis upwards. Successive snapshots of the body penetration are shown in Fig. 4.

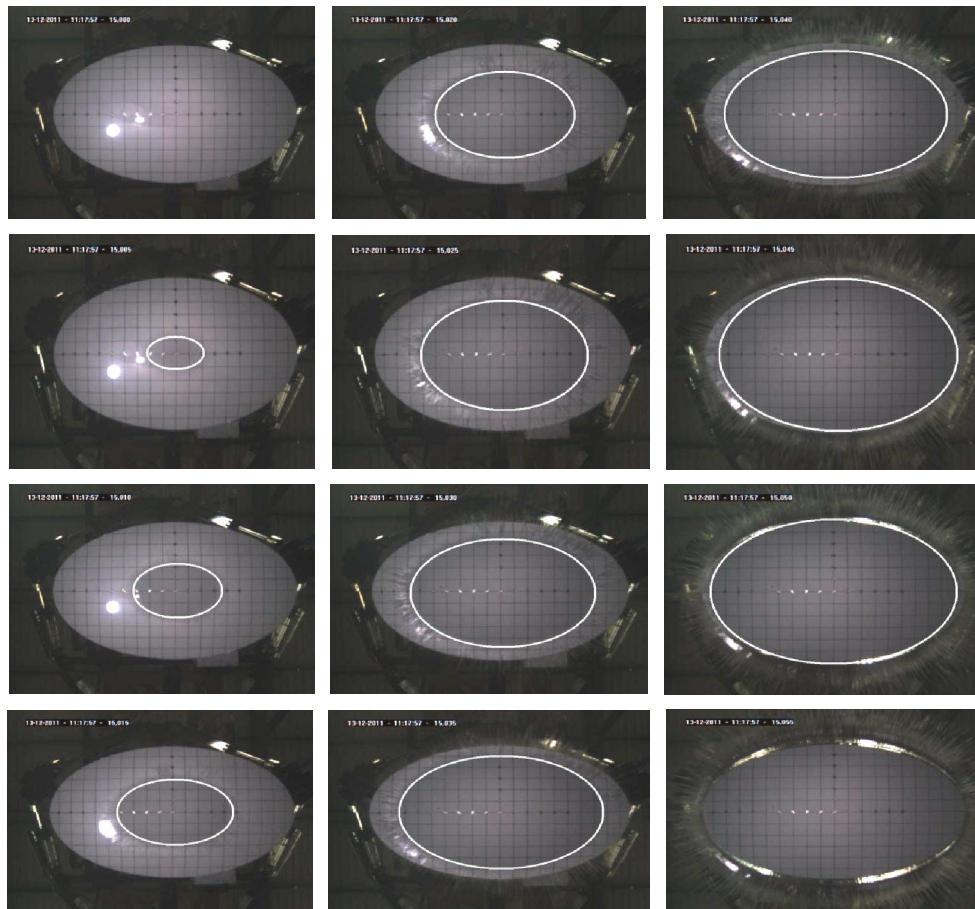


Fig. 4 Expansion of the wetted surface for the vertical and horizontal velocities components $W = 0.8600 \text{ m/s}$ and $V = 0.4650 \text{ m/s}$. The duration between each snapshot is $\Delta t = 0.005 \text{ sec}$.

The snapshots are ranged from top to bottom and from left to right. The limit of the wetted surface is underlined with the thick white lines.

From the image processing and by using the grid we detect the time variation of the expanding wetted surface. We collect the data of 12 tests for different couples of velocity components (V, W) . Fig. 5 illustrates the time variation of the wetted surface. The two lengths a and b are made non dimensioned with $W^{1/2}$ and they are plotted versus $t^{1/2}$. The theoretical variation is therefore linear.

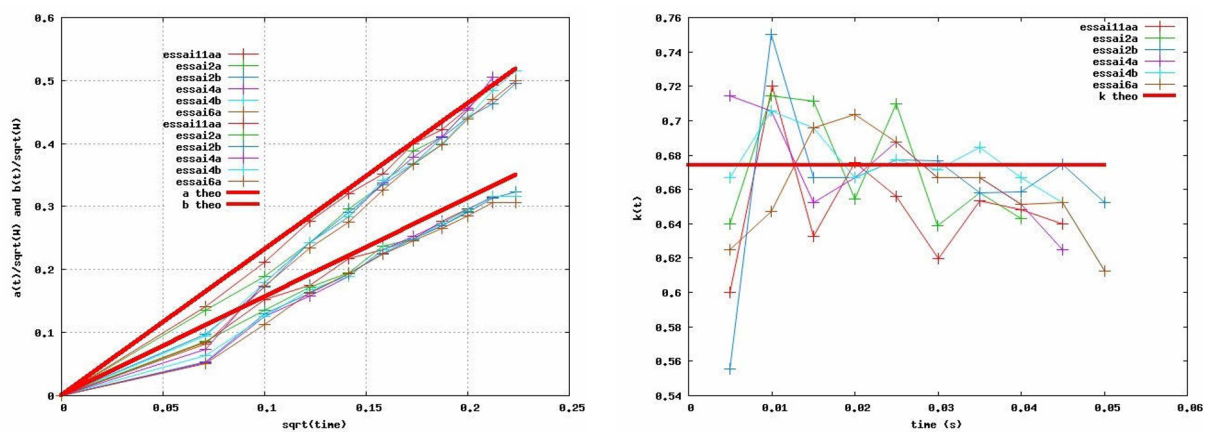


Fig. 5 (top): time variation of the length of the semi axes for different couple of velocities (V, W) . (bottom) time variation of the aspect ratio $k = a/b$.

The error between experiments and theory is within 10% except at the early stage of penetration when it is difficult to detect the contact line accurately. The absolute error of measurement is approximately one third the size of the cell grid (0.05 m) yielding the highest relative error of 20%.

Regarding the pressure we surprisingly get a rather satisfactory agreement. The pressure is measured at three and six points along the x and y directions respectively. The positions of the sensors are collected in the Table 1.

MLM pressure (13) is averaged over the sensor area. Here this area is circular with radius $r_{\text{sensor}} = 2.5 \text{ mm}$. A time lag is determined to adjust the theoretical and experimental pressures. Comparisons are shown in Fig. 6.

Table 1 Position and numbering of the pressure sensors along the main axes x and y.

Sensor	X (m)	Y (m)
1	0	0.1
2	0	-0.05
3	0	-0.15
4	0	-0.20
5	0	-0.35
6	0	-0.45
7	0.05	0
8	0.15	0
9	0.25	0

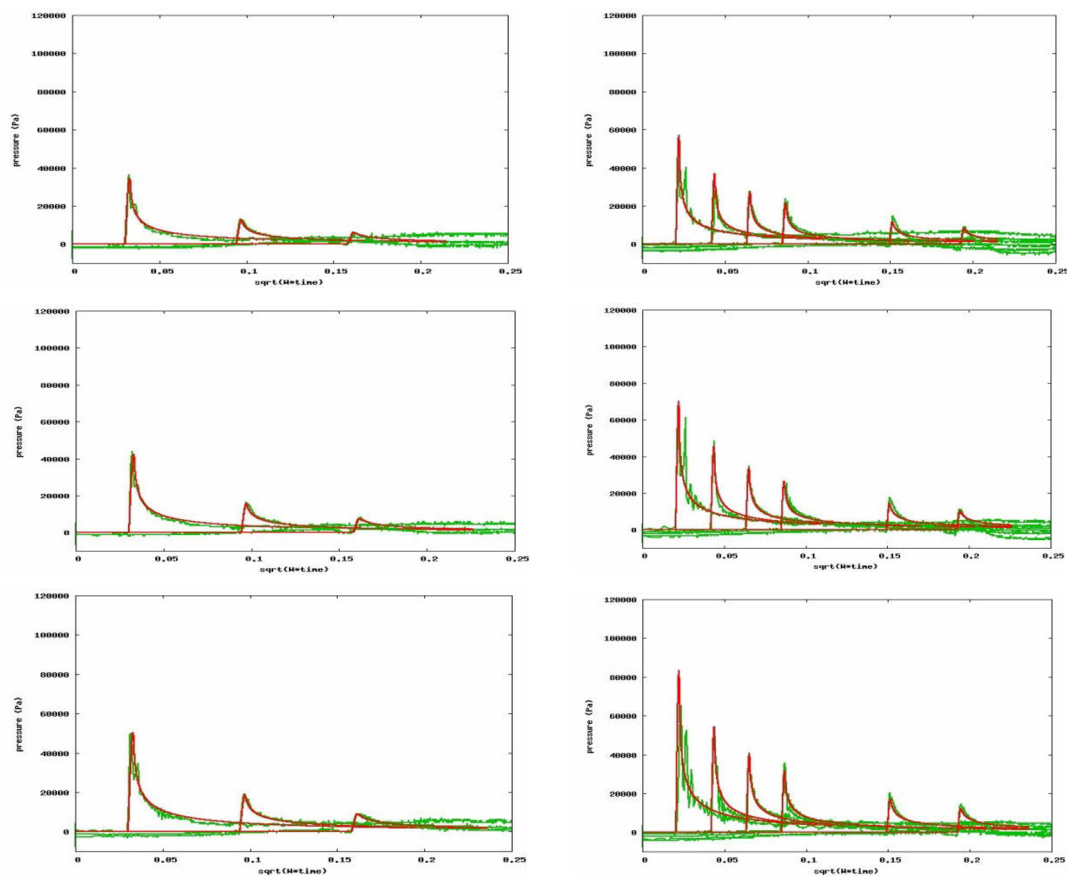


Fig. 6 Time variation of the pressure measured at the points for three couples of vertical and horizontal velocities: (top): $w = 0.7075 \text{ m/s}$, $v = 0.7175 \text{ m/s}$, (middle): $w = 0.7876 \text{ m/s}$, $v = 0.5850 \text{ m/s}$, (bottom): $w = 0.8575 \text{ m/s}$, $v = 0.4725 \text{ m/s}$.

Except for the first hit sensor, the agreement is rather good. As an example, we focus, in Fig. 7 on the couple of vertical and horizontal velocities ($W = 0.7876 \text{ m/s}$, $V = 0.5850 \text{ m/s}$) for which we plot the sudden rise up of the pressure when the pressure sensor is hit by the fluid.

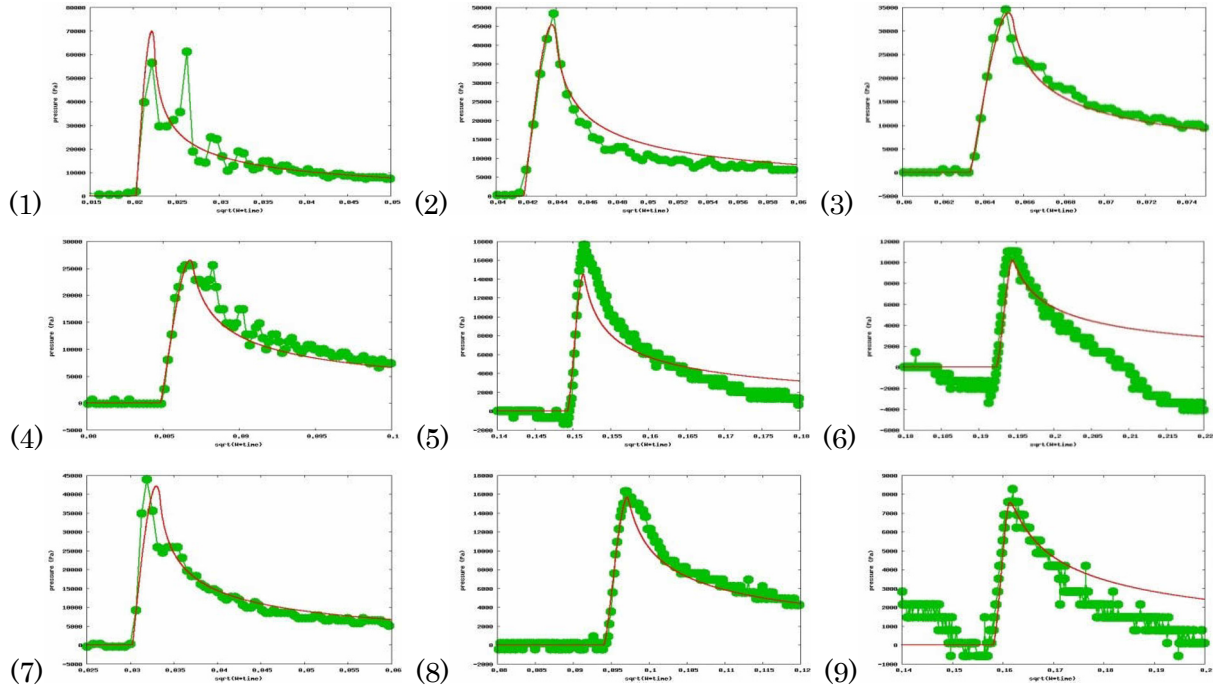


Fig. 7 Close view on the rise up of the pressure for the couple of vertical and horizontal velocities ($W = 0.7876 \text{ m/s}$, $V = 0.5850 \text{ m/s}$). The figures are numbered from top to bottom and from left to right; the numbers refer to the sensor number collected in Table 1.

It is clear that the only way to properly capture the rise up is the averaging of the pressure over the sensor area. The pressure is recorded with a sampling frequency of 20 KHz and it is not enough for the first hit sensors. However for the intermediate sensors, the rise up is well captured by the large enough number of points. For the last sensors, theoretical and measured data show large discrepancies. One of the reasons is the effects of the water suddenly ejected in the jet which provokes a depression before being hit by the bulk of the fluid. It should be noted that averaging the numerical pressure on the sensor area has also a strong influence on the maximum pressure. Here this maximum is rather well predicted while without averaging the maximum would be largely overpredicted as shown in Scolan (2014).

It should be noted that in the present experimental program, the horizontal velocity component is low (less than 1 m/s). Hence we cannot reach the time at which the pressure falls below the atmospheric pressure on the wetted surface. As shown in Moore et al. (2012) for a cone but also shown in Scolan and Korobkin (2014) for the present shape, the time at which the pressure falls below zero corresponds to $\dot{b} = \sqrt{2}V$ and that occurs along the axis $x = 0$ at

$$\tilde{y} = y - \delta y = -b/\sqrt{2}$$

If we assume that the two velocity components (horizontal and vertical) are both constant, this instant is

$$t_{p=0} = \frac{R_y W}{4V^2} \left(1 + k^2 \frac{D}{E} \right) \quad (19)$$

where D is a standard Integral Elliptic (see Appendix).

In the present case, $t_{p=0} = 0.9 \text{ sec.}$ for the couple of vertical and horizontal velocities $W = 0.71 \text{ m/s}$ and $V = 0.72 \text{ m/s}$. That

instant is much greater than the allowed duration of the Wagner stage. That would be always the case unless the horizontal velocity reaches high amplitudes. As an example the occurrence of zero pressure on the wetted surface will occur at time $t = 0.02 \text{ sec.}$ if the horizontal velocity is greater than $V = 5.8 \text{ m/s}$ while the vertical velocity is $W = 1 \text{ m/s}$.

CONCLUSION

The oblique entry of three dimensional body is analyzed in the frame of the Wagner theory. Comparisons with dedicated experimental data show that Wagner model is quite satisfactory not only in terms of the dynamics of the wetted surface but also in terms of the local loads. The comparisons regarding the global loads are less convincing. One of the reasons is the lack of stiffness of the experimental set-up. New experimental programs are highly required in order to confirm the role of the horizontal velocity in the appearance of low pressure areas.

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APPENDIX

Application of Galin's theorem

A formulation of Galin's theorem can be found in Vorovich et al. (1974), it reads:

Theorem: Let the functions (G, F, P, I) defined as follows:

$$G(x, y) = \frac{1}{\sqrt{x^2 + y^2}}, \quad F(x, y) = 1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}, \quad P(x, y) = \sum_{p=0}^M \sum_{q=0}^{M-p} b_{pq} x^p y^q \quad (20)$$

$$I(x, y) = \int_D \frac{P(x_0, y_0)}{\sqrt{F(x_0, y_0)}} G(x - x_0, y - y_0) dx_0 dy_0 \quad (21)$$

$$\text{if } (x, y) \in D = \left\{ (x, y), \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 \right\}, \quad \text{then } I(x, y) = \sum_{p=0}^M \sum_{q=0}^{M-p} a_{pq} x^p y^q \quad (22)$$

that means that I is a polynomial in (x, y) of the same degree as P .

A generalization of this theorem is given in Kalker (1973 and 1990). We assume in the sequel that Galin's theorem can be inverted: If I is polynomial in (x, y) , say with degree M , then P is polynomial as well with same degree M . In practice knowing the coefficients a_{pq} , we can determine explicitly the coefficients b_{pq} as shown in Vorovich et al. (1974).

We consider the elliptic paraboloid whose shape function is given by

$$f(x, y) = \frac{x^2}{A^2} + \frac{y^2}{B^2} \quad (23)$$

where (A^2, B^2) are twice the curvature radii at the initial contact point, denoted (R_x, R_y) . The RHS of (2) is a polynomial of degree 2. The two integral Eqs. (2) and (21) are identical and Galin's theorem, applies. Then the planar Laplacian S in (2) can be written as

$$S(x, y) = \frac{1}{\sqrt{F}} (b_{00} + b_{20}x^2 + b_{02}y^2) \quad (24)$$

where (b_{00}, b_{20}, b_{02}) are unknown coefficients at that stage. Symmetries of the geometry and solutions impose to retain even polynomials only. The identification of each coefficient of 1, x^2 and y^2 on both hands of Eq. (2) leads to a linear system

$$\begin{pmatrix} S_{11} & S_{12} & S_{13} \\ 0 & S_{22} & S_{23} \\ 0 & S_{32} & S_{33} \end{pmatrix} \begin{pmatrix} b_{00} \\ b_{20} \\ b_{02} \end{pmatrix} = \begin{pmatrix} 2h \\ -\frac{2}{A^2} \\ -\frac{2}{B^2} \end{pmatrix} \quad (25)$$

where $a = kb$, $k^2 = 1 - e^2$ and s_{ij} are known functions of eccentricity e in terms of standard Elliptic Integral functions. The linear system (25) can be inverted so that (b_{00}, b_{20}, b_{02}) are calculated in terms of (A, B, h, a, b) . However it should be noted that (a, b) are still unknown. The expression of the function S suggests that the displacement potential itself reads

$$\phi(x, y, t) = H(x, y, t) \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} \right)^{3/2} \quad (26)$$

where function H is an unknown function at the present stage. Some elements of proof of that result can be found in Sclan and Korobkin (2008; 2012b). The boundary conditions (3) and (4) on $\Gamma(t)$ are hence implicitly accounted for. By taking the spatial derivatives of ϕ , and arriving at the planar Laplacian, we conclude that H is a constant function of time t only, hence H does not depend on spatial coordinates (x, y) . We can then express the planar Laplacian of the displacement potential

$$\Delta_2 \phi(x, y, t) = \frac{3H}{\sqrt{F}} \left(-\frac{1}{a^2} - \frac{1}{b^2} + \frac{x^2}{a^2} \left(\frac{2}{a^2} + \frac{1}{b^2} \right) + \frac{y^2}{b^2} \left(\frac{1}{a^2} + \frac{2}{b^2} \right) \right) \quad (27)$$

Identifying (27) and (24) lead to three additional equations to already established Eq. (25).

$$b_{00} = -3H \left(\frac{1}{a^2} + \frac{1}{b^2} \right), \quad b_{20} = \frac{3H}{a^2} \left(\frac{2}{a^2} + \frac{1}{b^2} \right), \quad b_{02} = \frac{3H}{b^2} \left(\frac{1}{a^2} + \frac{2}{b^2} \right) \quad (28)$$

That is enough to obtain (b_{00}, b_{20}, b_{02}) , (a, b) and H in terms of (A, B, h)

$$H = -\frac{2ha}{3E}, \quad b = B \sqrt{h \left(1 + k^2 \frac{D}{E} \right)} = \eta \sqrt{h}, \quad a = A \sqrt{h \left(2 - k^2 \frac{D}{E} \right)} \quad (29)$$

Where $K(e)$, $E(e)$ and $D(e)$, $(K(e) - E(e)) / e^2$ are the standard Elliptic Integrals (see Gradshteyn and Ryzhik, 2007, p860). From these last equations, we get

$$k_\gamma^2 = \frac{A^2}{B^2} = k^2 \frac{1 + k^2 D/E}{2 - k^2 D/E} \quad (30)$$

which are exactly the equations (6.13) or (6.22) of Sclan and Korobkin (2001). Fig. 8 shows that the function k_γ is a function one to one in terms of k .

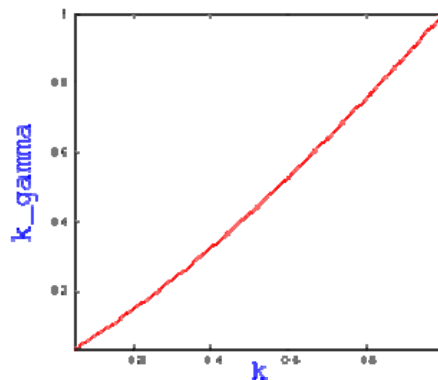


Fig. 8 Variation of k_γ in terms of k according to Eq. (30).